WSU AIRPACT UPDATE

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with contribution from:
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NW-AIRQUEST Consortium Meeting, February 6, 2026

Budget Status

FY26 Budget

- **Salaries,** Tuition and Benefits = $105k
  - Graduate Research Assistants
  - Recruiting a new Postdoc
  - Recruiting a new PhD student
- **Web maintenance and other service** = $12k
- **HPC Equipment** = $30k
- **WSU overhead** (12%) = $13k
- **TOTAL = $160,000**

Budget Status

- FY26 is the second year of a two-year (biennium) budget
- The award is still processing at WSU due to the initial delay and understaffed WSU ORSO office

Equipment

Equipment update

- The **Aeolus HPC** cluster continues to run the AIRPACT5 modeling system each night – Aeolus HPC is approaching decommissioning
- **GAIA**: 32-thread Linux workstation is being used for development and testing
- The new **Kamiak HPC** cluster has 3 new computing nodes (192 processers) with 1.5 TB of total RAM for LAR; we are purchasing 2 new computing nodes to be implemented on Kamiak.

On-going Tasks

ONGOING TASKS

(Courtesy of Jahidul)

Task	Status June 2025	
1. Provide daily forecasts of air quality for the three-state region that can be used in daily AQ management.	No issue	

ONGOING TASKS

Task	Status June 2025	
2. Provide near real-time verification statistics of the current AIRPACT system accuracy by pollutant, by month, and Provide for user access to AIRPACT results and monitoring data for user-defined periods for download & analysis.	Statistics continue to be maintained.	

ONGOING TASKS

Task	Status June 2025	
3. Track and continue to improve model performance so that model predictions provide useful guidance for air quality forecasting.	Continuing; no issues.	

ONGOING TASKS

Task	Status June 2025	
4. Maintain ambient air monitoring data streams from AIRNow.	Continuing; no issues (Also have Python scripts for easy download upon request)	
5. Maintain satellite retrievals that are still functioning (except for MODIS and OMI).	Satellite data are no longer being actively archived	
6. Continue improvements and requested additions to web graphical		

ONGOING TASKS

Task	Status June 2025	
7. Maintain the “Change Log”, posted on the AIRPACT website. 8. Archive daily model output of the first 24 hours. Web-based graphical displays for each day’s first 24 hours of forecast should be available for the most recent 5 years & gridded files for at least one month.	Ongoing. Now stored at GitHub; https://github.com/wsular/airpact Ongoing. Forecast graphics available beginning December 2015.	

ONGOING TASKS

Task	Status June 2025	
9. Modify AIRPACT as needed to respond to changes in fire emissions produced by other organizations. 10. Continue hosting the link to the latest background concentration lookup tool.	Ongoing as needed Hosted by IDEQ; Link will be added to AIRPACT site under Related Programs -> Graphics and Data	

ONGOING TASKS

Task	Status June 2025	
11. Modeling infrastructure maintenance and updates.	$30,000 investment in FY26 two more HPC nodes (64 processes each; 128 total) in WSU Kamiak HPC)	

FY26 Tasks

FY26 PROPOSED TASKS

1. AIRPACT6 on Kamiak HPC

- MCIP v5.5 to process UW WRF model output on Kamiak HPC
- CMAQ model version upgrades to v5.5 (the latest version)
- **SMOKE v5.2 with EPA Emission Modeling Platform (EMP) 2022**
  - **12 km US emission for CB06 and CRACMM2**
  - **4km AIRPACT domain emission for CB06 and CRACMM2**
- Chemical Mechanisms
  - CB06
  - CRACMM (Pye et al., 2023)



Ongoing task, Priom will present the details

FY26 PROPOSED TASKS

Ongoing task

2. Comprehensive AIRPACT6 Validation

- Compare with AIRPACT5 for concurrent period (summer month and winter month)
- Evaluate with ground-based measurements (AQS)
- Evaluate with other satellite-derived products
- Summarize the differences between two different chemical mechanisms

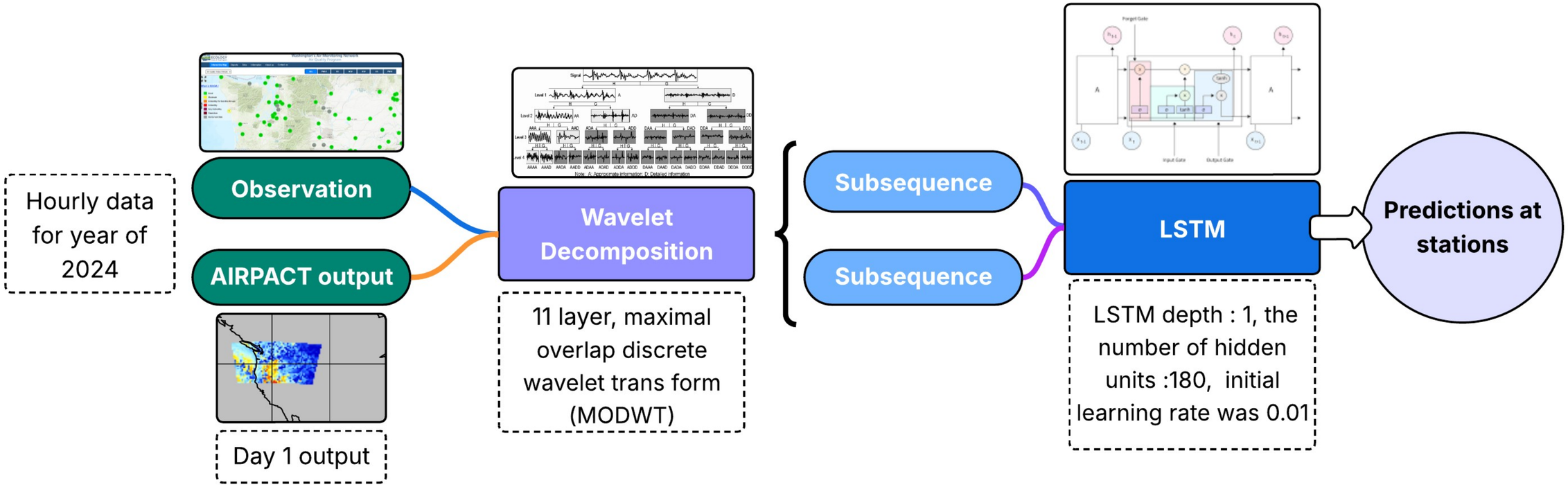
FY26 PROPOSED TASKS

We will present the final product in June meeting

3. New Machine Learning Forecasting Model

- for monitoring sites and spatial map forecasting

Hourly Forecast Model Framework



(Courtesy of Priom Zarrah)

Summary

Summary

- AIRPACT6 is being tested on a local workstation and being migrated to a new HPC
- A new ML model is very promising for AQ forecasting (monitoring sites and spatial)
- FY26 proposed tasks are undergoing with huge progress on emission development
- Monthly modeling meeting has been super helpful; next meeting will be in late feb.

Thank You!
Any questions?

